

Facebook Network Analysis Report

Dataset Information

The dataset used in this project is the SNAP Facebook Social Circles dataset provided by the Stanford Network Analysis Project (SNAP). The file used for the analysis was:

facebook_combined.txt

The dataset contains friendship connections between Facebook users represented as edges in a social network graph.

The dataset was converted into CSV format (facebook_edges.csv) and imported into Gephi as an undirected graph for analysis.

Network Overview

After importing the dataset into Gephi, the following network statistics were obtained:

Metric	Value
Number of Nodes	4039
Number of Edges	88234
Graph Type	Undirected

The network is dense and highly interconnected, which is common in social media friendship networks.

Network Visualization

The ForceAtlas2 layout algorithm was applied to organize the graph visually. This layout helped separate clusters and reveal the overall structure of the network.

Node sizes were adjusted according to degree centrality, where highly connected nodes appeared larger.

Two visualizations were exported:

- Full network visualization
- Community/modularity visualization

Degree Centrality Analysis

Degree centrality measures how many direct connections each node has in the network. Nodes with high degree centrality act as hubs because they are connected to many other users.

Top 5 Nodes by Degree Centrality

Node ID	Observation
107	Highest connectivity in the network
1684	Major hub node
1912	Strongly connected user
3437	Important highly connected node
0	Central participant in the network

These nodes are important because they connect many users and help maintain the network structure.

Betweenness Centrality Analysis

Betweenness centrality identifies nodes that act as bridges between communities. These nodes help connect different groups and are important for information flow across the network.

Top 5 Nodes by Betweenness Centrality

Node ID	Observation
107	Major bridge between communities
1684	Connects multiple clusters
1912	Important intermediary node
3437	Connects social groups
0	Influential connector

Nodes with high betweenness centrality may not always have the highest degree, but they play an essential role in connecting separate communities.

Community Detection

The Modularity algorithm was used to detect communities within the Facebook network.

Different colors were assigned to different modularity classes in Gephi. The analysis revealed multiple densely connected communities, representing friend groups or social circles.

The modularity visualization clearly showed:

- Strong clustering behavior
- Separation between communities
- Dense internal connections within groups

This demonstrates how social networks naturally form clusters of users with similar connections or interests.

Degree Filter Analysis

A Degree Range filter was applied with:

Degree ≥ 5

This removed weakly connected nodes from the visualization.

Comparison Between Original and Filtered Network

Original Network

- Large and dense
- Included peripheral nodes
- More difficult to interpret visually

Filtered Network

- Cleaner structure
- Important communities became more visible
- Central hub nodes were easier to identify
- Reduced visual clutter

The filtering process improved readability and highlighted the core structure of the network.

Interpretation of the Network Structure

The Facebook social network shows characteristics of a real-world social network:

- Presence of highly connected hub nodes
- Multiple dense communities
- Bridge nodes connecting clusters
- Strong clustering and social grouping behavior

The network follows a scale-free structure where a small number of nodes have many connections while most nodes have relatively few.

The analysis also demonstrated that social networks are not random. Instead, users organize into communities connected through influential individuals.

Learning Outcomes

From this analysis, I learned:

- How to import and analyze network datasets in Gephi
- How to use layouts such as ForceAtlas2 for graph visualization
- How centrality measures identify influential nodes
- How betweenness centrality detects bridge nodes
- How modularity reveals communities in networks
- How graph filtering improves network readability and interpretation

This project provided practical experience in social network analysis and visualization using Gephi.